

MECHANICAL DESIGN PORTFOLIO



SIVA SAI KAMMARA

MECHANICAL DESIGN ENGINEER

SOLIDWORKS

Kadapa, Andhra Pradesh, India

Phone: +91 7013029502

Email: sivasaik9999@gmail.com

LinkedIn: [linkedin.com/in/siva-sai-kammara-sk9999](https://www.linkedin.com/in/siva-sai-kammara-sk9999)



PROJECT - 02

SHOCK ABSORBER ASSEMBLY

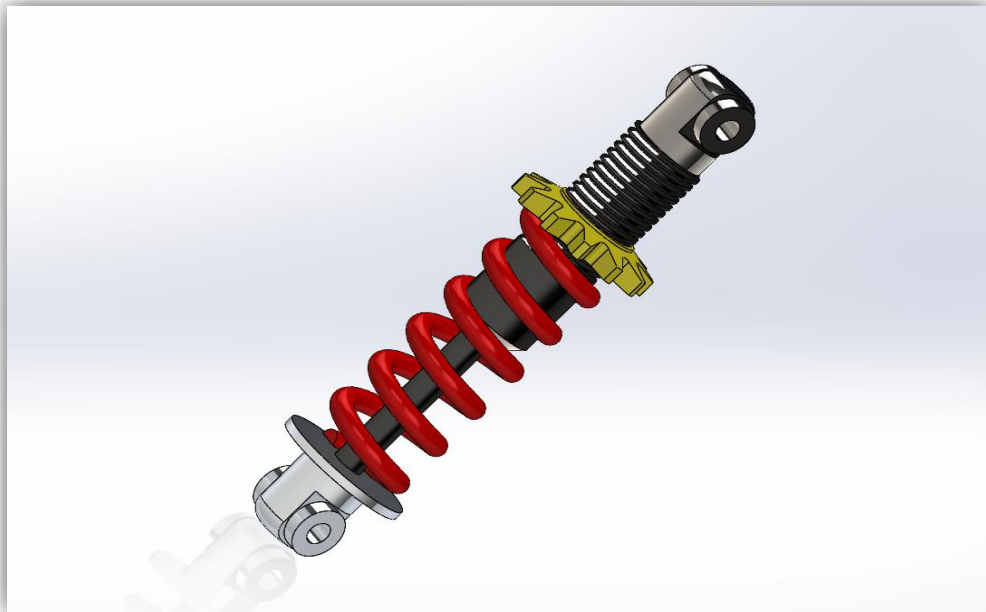


Fig. 1 – Shock Absorber Assembly

Complete assembled model of the shock absorber created in SolidWorks.

SOFTWARE USED

SolidWorks 2023

PROJECT TYPE

3D Part Modeling and Assembly

OBJECTIVE

Designed and developed a complete 3D Shock Absorber Assembly using SolidWorks by modeling individual components and assembling them to enhance practical mechanical design, CAD modeling, and assembly skills.

COMPONENTS MODELED

- Lower Mount
- Piston Rod
- Helical Compression Spring
- Lock Nut

WORK PERFORMED

- Created accurate 3D models of all shock absorber components.

- Assembled the complete shock absorber using appropriate assembly mates.
- Verified proper alignment and fit of all assembled parts.
- Designed the helical compression spring using the Sweep feature.
- Created an exploded view of the assembly to illustrate the arrangement and relationship of individual components.
- Applied mechanical design and assembly techniques to develop a functional shock absorber assembly.

CAD FEATURES USED

- Sketch
- Extrude Boss/Base
- Extrude Cut
- Revolve
- Sweep Boss/Base
- Helix and Spiral
- Fillet
- Chamfer
- Mirror
- Circular Pattern
- Assembly Mates
- Exploded View

SKILLS DEMONSTRATED

- 3D Part Modeling
- Assembly Modeling
- Mechanical Design
- Suspension System Design
- SolidWorks
- Engineering Drawing Basics

PROJECT GALLERY



Fig. 2 – Piston Rod with Lower Mount
Connects the shock absorber to the



Fig. 3 – Shock Absorber Body
Provides damping for the suspension.

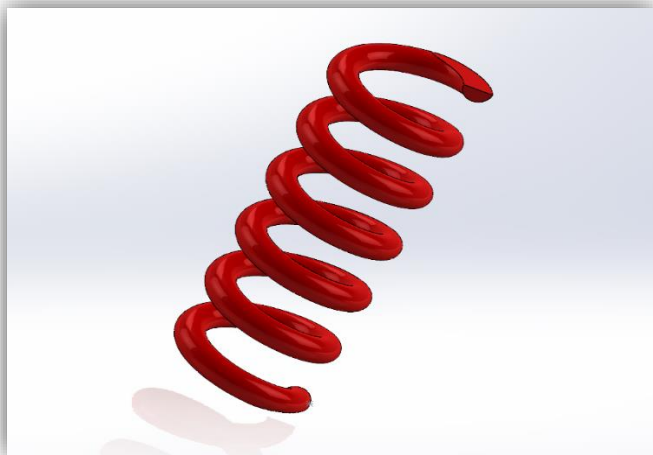


Fig. 4 – Helical Compression Spring
Absorbs road shocks and supports the vehicle load.

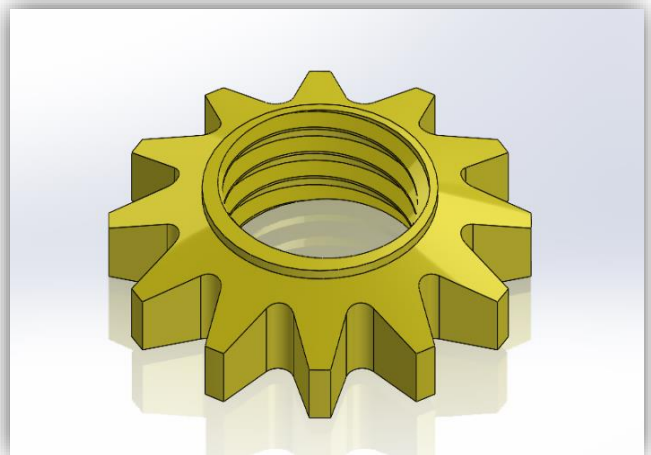


Fig. 5 – Preload Adjusting Lock Nut
Adjusts and secures the spring preload.

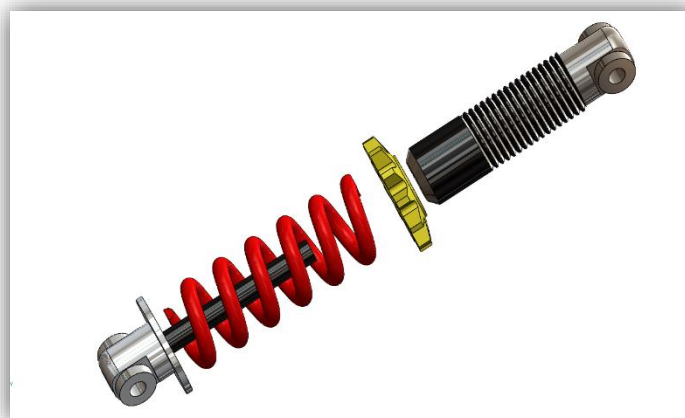


Fig. 6 – Shock Absorber Exploded View
Displays the individual components of the shock absorber before assembly.

“SCAN TO WATCH THE SHOCK ABSORBER EXPLODED VIEW ANIMATION”

